

Balaji Ramachandran

Education And Achievements:

Carnegie Mellon University <i>PhD Student In Mathematical Sciences</i>	Pittsburgh 2026-
Indian Institute Of Science(IISc) <i>Bachelors in Technology In Mathematics And Computing, CGPA-9.2/10.0</i> Entrance Exam: JEE Advanced(All India Rank 225)	Bangalore 2022-2026

ML Projects:

Stochastic Path Planning for Autonomous Underwater Vehicles:

- Solved Hamilton–Jacobi–Bellman equations for optimal navigation in stochastic ocean flows
- Obtained optimal trajectories with decision transformer trained on the solution of the HJB equations.
- This project introduced me to Scientific Machine Learning

AI based Market Research Analyst:

- Developed a web application that automates market-research workflows using Firebase, LangChain, and BeautifulSoup selected as the best project for presentation at a GenAI symposium held at IISc.

LLM-Based Tutoring System: Applied Monte Carlo Tree Search to finetuned large language models for mathematical reasoning tasks [Medium Article](#)

Skills

AI For Science: Experience with state of the art techniques for SciML such as PINNs, Diffusion and Flow Based Models

Machine Learning: Reinforcement Learning, Optimization, Generative AI, Optimal Transport, Statistical Learning

Scientific Computing: Numerical PDEs, HPC, Monte Carlo Simulations

Languages: Python, C++

Research Experience:

Research Intern, Multi-Agent Systems Lab, IISc: Advisor: Prof. Pavankumar Tallapragada

- Studied network formation games under budget constraints
- Analyzed Nash equilibrium network topologies arising from strategic weight allocation among agents looking to maximize their centralities
- Paper accepted at CDC 2026 [Paper link](#)

Summer Research Intern, Stochastic Systems Lab, Purdue IEOR: Advisor: Prof. Harsha Honnappa

- Developed reinforcement learning–based methods for stochastic impulse control of diffusion

processes

- Solved associated PDEs using Physics Informed Neural Networks
- Worked on non-parametric drift estimation for diffusions using techniques inspired by score matching

Undergrad Researcher, Risk Lab, IISc: Advisors: Prof. Srikanth Iyer, Prof. Shashi Jain

- Formulating a Markov Decision Process model for optimal treasury allocation under interest-rate risk
- Focus on dynamic rebalancing policies and long-horizon risk management in banking portfolios

Analyst Intern, HSBC: Worked on the Stress scenario risk management dashboard for the Traded Risk Analytics team from January to June 2026

Relevant Coursework

- **Machine Learning:** Reinforcement Learning, AI For Scientific Computing, Scalable Systems, Applied Data Science
- **Mathematics and Statistics:** Probability Theory, Optimal Transport, Stochastic Processes, Regression and Time Series Analysis, Nonlinear Systems and Control, Numerical Methods

Additional Course Projects

- Implemented deterministic actor–critic algorithms; explored TRPO, TD learning, and Q-learning
- Presented a term paper on diffusion models(DDPMs and implementation)
- Used PySpark and HPC for experiments on large datasets
- Presented the convergence analysis of the Langevin Monte Carlo algorithm used in generative modeling and optimization
- Applied ARMA/ARIMA models for financial time series